Picking the Next Token, Greedy Decoding to Beam Search

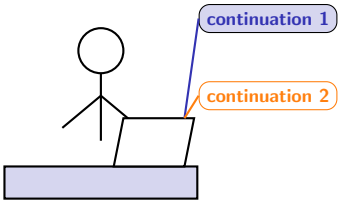
Text Analytics, Chapter 7

Prof. Dr. Jörg Osterrieder

One Prompt, Two Continuations

Scenario: You send one prompt to a chatbot twice and receive two different continuations.

- After the prompt, the language model has a probability p_i for each token i of its vocabulary V
- One token is appended, and the step is repeated on the longer text
- With a random draw at each step, the texts of two runs differ
- The question: which token is appended, under which rule



The appended token depends on the decoding rule.

Learning Goals for Decoding

- **1 Next-Token Distribution**
  Compute next-token probabilities from scores; write the probability of a continuation as a product.

2 Greedy Decoding
Name the greedy token; explain the repetition loop.

3 Temperature
Name the limits of $p^{(T)}$ as $T \to 0$, $T \to \infty$.

4 Top-k, Top-p
Choose top-k or top-p for a peaked or a flat distribution.

5 Beam Search
Run beam search of width B on a two-step tree.

6 Worked Example: GPT-2 Small
Compute each rule on the numbers of GPT-2 small after one prompt.

▶ next under each part ahead: its learning goal

Worked Example: GPT-2 small after the first words of *Pride and Prejudice*.

Softmax Probabilities From Scores

Definition

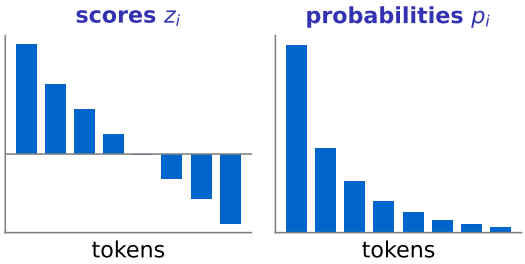
After a context c, the language model has a **score** z_i for each token i of the vocabulary V. The **softmax** of the scores is the next-token probability

$$p_i = \frac{e^{z_i}}{\sum_{j \in V} e^{z_j}}.$$

Intuition:

- A larger score $\rightarrow$ a larger probability
- $p_i > 0$ for every token, $\sum_i p_i = 1$
- $z_i + a$ for every token $\rightarrow$ the same p: p depends only on differences of scores

Schematic: softmax of the scores



The softmax of the scores is the next-token distribution p.

Probability of a Continuation as a Product

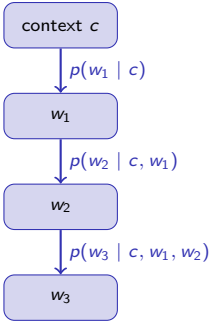
Definition

A **continuation** $w_1, \ldots, w_n$ of the context c has the probability

$$P = \prod_{t=1}^{n} p(w_t \mid c, w_{<t}), \quad \log P = \sum_{t=1}^{n} \log p(w_t \mid c, w_{<t}).$$

Intuition:

- Each factor is at most $1 \rightarrow P$ is smaller with every token
- $\log P \leq 0$; closer to $0 \rightarrow$ a more probable continuation
- A large first factor can be followed by small ones



$$P = p(w_1 \mid c)\, p(w_2 \mid c, w_1)\, p(w_3 \mid c, w_1, w_2)$$

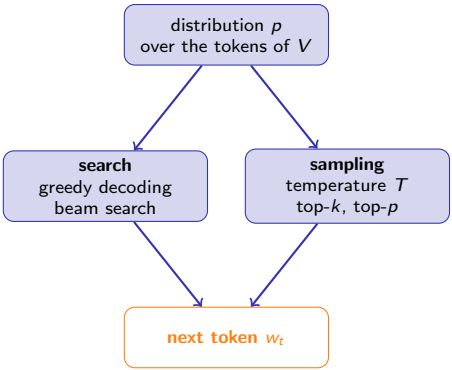
The probability of a continuation is the product of its next-token probabilities.

Decoding Rules: Search, Sampling

Definition

A **decoding rule** is a map from the distribution p at a step to the next token: by a **search** for a probable continuation, or by a **draw** from a distribution q derived from p.

- Search: greedy decoding, beam search → one text for one prompt
- Sampling: temperature T, top-k, top-p → a new text on each run
- Temperature first, then a cut by top-k or top-p: the sampling rules are combined



Under search, one prompt gives one text; under sampling, each run gives a new text.

Greedy Decoding

✓ 1 Next-Token Distribution

▶ **2 Greedy Decoding**
Name the greedy token; explain the repetition loop.

3 Temperature
Name the limits of $p^{(T)}$ as $T \to 0$, $T \to \infty$.

4 Top-k, Top-p
Choose top-k or top-p for a peaked or a flat distribution.

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✓ finished ▶ next under each part ahead: its learning goal

The Greedy Token Has the Largest Probability

Definition

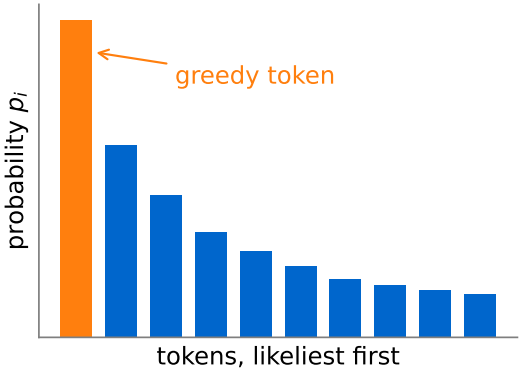
Under **greedy decoding**, the next token is the token with the largest probability, at every step:

$$w_t = \arg\max_{i \in V} p(i \mid c, w_{<t}).$$

Intuition:

- No draw $\rightarrow$ one prompt, one text, on every run
- Each step is decided alone: later probabilities are left out of the comparison
- Greedy decoding is temperature at $T \rightarrow 0$ and beam search at $B = 1$

Schematic: greedy token in orange



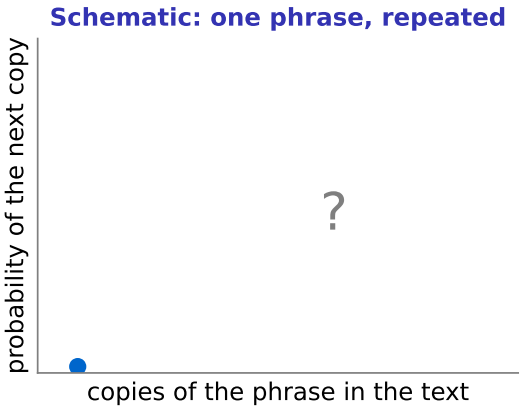
Under greedy decoding, the likeliest token is appended at every step.

Probability of a Repeated Phrase

Under greedy decoding, the end of a text can be a loop: one phrase, repeated again and again.

Discovery

Each copy of the phrase is added to the context. With every copy, does the probability of the next copy grow or fall?



Under greedy decoding, the probability of a repeated phrase grows with each copy.

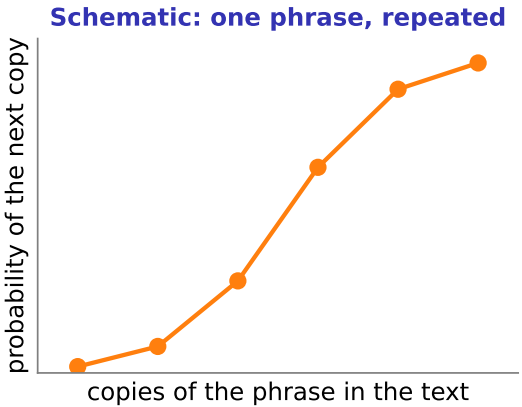
Probability of a Repeated Phrase

Under greedy decoding, the end of a text can be a loop: one phrase, repeated again and again.

Discovery

Each copy of the phrase is added to the context. With every copy, does the probability of the next copy grow or fall?

Answer: it grows. With each copy in the context, the next copy has a larger probability, and the loop is unbroken to the last token (Holtzman et al., 2020).



Under greedy decoding, the probability of a repeated phrase grows with each copy.

Temperature

✓ 1 Next-Token Distribution

✓ 2 Greedy Decoding

▶ **3 Temperature**
Name the limits of $p^{(T)}$ as $T \to 0$, $T \to \infty$.

4 Top-k, Top-p
Choose top-k or top-p for a peaked or a flat distribution.

5 Beam Search
Run beam search of width B on a two-step tree.

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✓ finished ▶ next under each part ahead: its learning goal

Scores Divided by T Before the Softmax

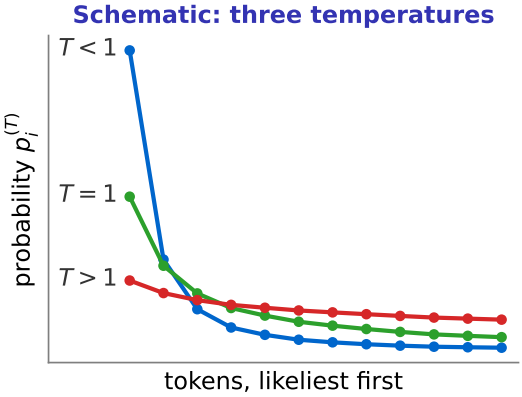
Definition

Under **sampling**, the next token is drawn at random from a distribution q. With **temperature** $T > 0$, $q = p^{(T)}$:

$$p_i^{(T)} = \frac{e^{z_i / T}}{\sum_{j \in V} e^{z_j / T}}.$$

Intuition:

- $T < 1 \rightarrow$ more mass on the likeliest tokens
- $T > 1 \rightarrow$ the mass spread over more tokens
- The order of the tokens is the same at every T; $T = 1$ gives p



Under temperature T, the scores are divided by T before the softmax.

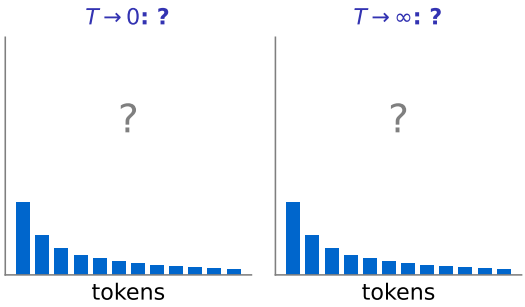
Two Limits of the Temperature

Temperature T in the softmax: $p_i^{(T)} \propto e^{z_i/T}$, with two limits.

Discovery

Which distribution is $p^{(T)}$ as $T \to 0$, which as $T \to \infty$?

Schematic: $p^{(T)}$ at $T = 1$



At $T \to 0$, sampling equals greedy decoding; at $T \to \infty$, every token has probability $1/|V|$.

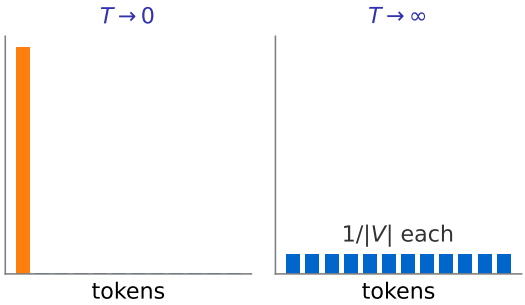
Two Limits of the Temperature

Temperature T in the softmax: $p_i^{(T)} \propto e^{z_i/T}$, with two limits.

Discovery

Which distribution is $p^{(T)}$ as $T \to 0$, which as $T \to \infty$?
Answer: $T \to 0$: all mass on the largest score, so a draw from $p^{(T)}$ equals greedy decoding. $T \to \infty$: $z_i/T \to 0$ for every token, so $p_i^{(T)} \to 1/|V|$, the uniform distribution.

Schematic: two limits of $p^{(T)}$



At $T \to 0$, sampling equals greedy decoding; at $T \to \infty$, every token has probability $1/|V|$.

Entropy Grows With Temperature

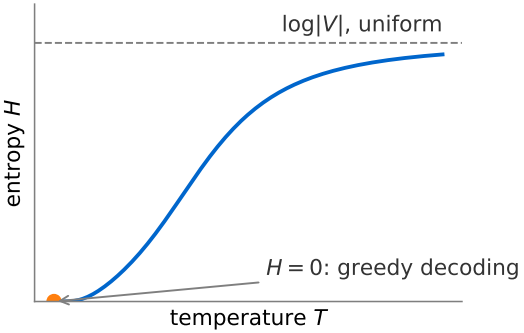
Definition

The **entropy** of a distribution p is $H(p) = -\sum_i p_i \log p_i$, in nats. $H = 0$ when one token has all the mass; $H = \log |V|$ under the uniform distribution.

Intuition:

- $\frac{dH}{dT} = \frac{\mathrm{Var}_{p^{(T)}}(z)}{T^3} \geq 0$ (Appendix A2)
- H grows with T, from 0 to $\log |V|$
- A larger $H \rightarrow$ a less predictable next token

Schematic: entropy against T



The entropy of $p^{(T)}$ grows with the temperature, from 0 to $\log |V|$.

Top-k, Top-p

✓ 1 Next-Token Distribution

✓ 2 Greedy Decoding

✓ 3 Temperature

▶ **4 Top-k, Top-p**
Choose top-k or top-p for a peaked or a flat distribution.

5 Beam Search
Run beam search of width B on a two-step tree.

6 Worked Example: GPT-2 Small
Compute each rule on the numbers of GPT-2 small after one prompt.

✓ finished ▶ next under each part ahead: its learning goal

Top-k: the k Likeliest Tokens

Definition

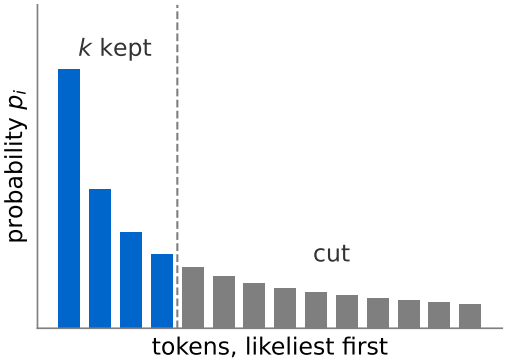
Under **top-k** sampling, the set S_k of the k likeliest tokens is kept, with the mass $M_k = \sum_{i \in S_k} p_i$. The draw is from

$$q_i = p_i / M_k \quad (i \in S_k), \qquad q_i = 0 \text{ otherwise.}$$

Intuition:

- The number of kept tokens is fixed at k
- The kept mass M_k depends on p
- $k = 1 \rightarrow$ greedy decoding; $k = |V| \rightarrow p$ itself

Schematic: top-k, kept tokens in blue



Under top-k, the k likeliest tokens are kept, whatever their mass.

Top-p: the Smallest Set With Mass π

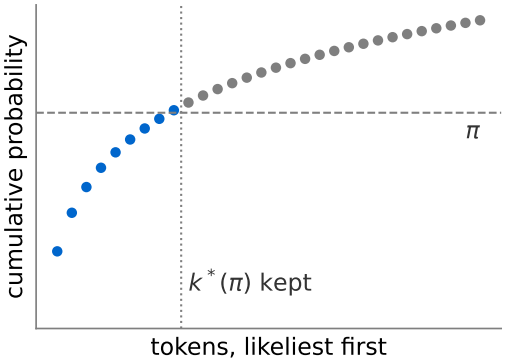
Definition

Under **top-p** sampling, or nucleus sampling, the kept set is the smallest set of likeliest tokens with mass $M \geq \pi$. The draw is from $q_i = p_i/M$ on that set; its size is $k^*(\pi)$.

Intuition:

- The kept mass is fixed: $M \geq \pi$
- The number of kept tokens $k^*(\pi)$ depends on p
- $\pi \to 0 \to$ greedy decoding; $\pi = 1 \to p$ itself

Schematic: top-p, kept tokens in blue



Under top-p, the likeliest tokens are kept until their mass is at least π.

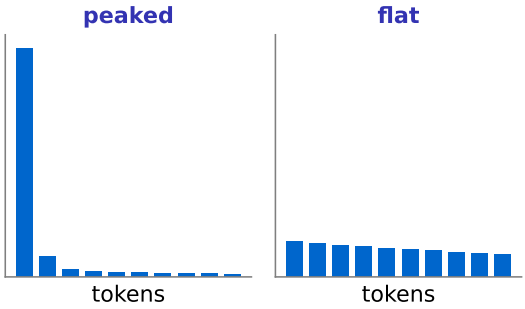
Fixed k Against Fixed π on Two Distributions

Two distributions: a peaked one, with most of the mass on one token; a flat one, with the mass spread over many tokens.

Discovery

One k, one π for both. Under top-k, after which distribution is the cut mass $1 - M_k$ larger? Under top-p, after which are more tokens kept?

Schematic: two distributions



Under top-k the number of kept tokens is fixed; under top-p, the kept mass.

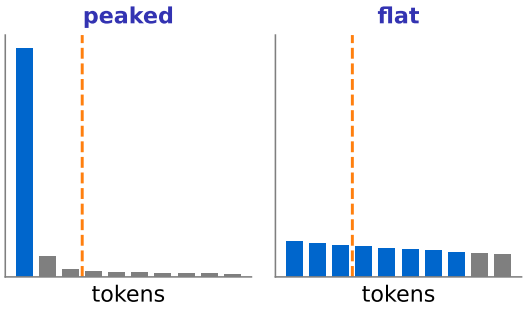
Fixed k Against Fixed π on Two Distributions

Two distributions: a peaked one, with most of the mass on one token; a flat one, with the mass spread over many tokens.

Discovery

One k, one π for both. Under top-k, after which distribution is the cut mass $1 - M_k$ larger? Under top-p, after which are more tokens kept?
Answer: the flat one, both times. Under top-k, most of its mass is cut, and almost none of the peaked one. Under top-p, the kept mass is at least π after both, with many tokens of the flat one, few of the peaked one.

Schematic: dashed k; blue top-p set



Under top-k the number of kept tokens is fixed; under top-p, the kept mass.

Beam Search

- ✓ 1 Next-Token Distribution
- ✓ 2 Greedy Decoding
- ✓ 3 Temperature
- ✓ 4 Top-k, Top-p
- ▶ **5 Beam Search**
  Run beam search of width B on a two-step tree.
- 6 Worked Example: GPT-2 Small
  Compute each rule on the numbers of GPT-2 small after one prompt.

✓ finished ▶ next under each part ahead: its learning goal

Beam Search of Width B in Four Steps

Definition

Under **beam search** of width B, the B partial continuations with the largest $\log P$ are kept at every step; the result has the largest $\log P$ at the end.

The four steps:

1. **Start** with the context c as one partial continuation, $\log P = 0$
2. **Extend** each kept partial continuation by every token i: $\log P + \log p(i \mid \cdot)$
3. **Keep** the B largest of the $B \cdot |V|$ extensions
4. **Repeat** steps 2, 3 until n tokens are appended

Discovery

At $B = 1$, which decoding rule is beam search?

Beam Search of Width B in Four Steps

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Discovery

At $B = 1$, which decoding rule is beam search?
Greedy decoding: one partial continuation, extended by its likeliest token.

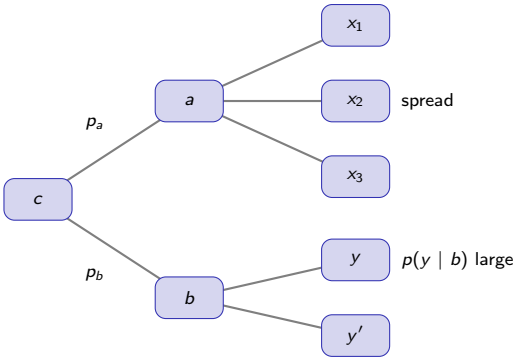
Beam search of width 1 is greedy decoding.

Greedy Path Against the Most Probable Path

Two first tokens a, b with $p_a > p_b$. After a, the mass is spread over many tokens; after b, most of it is on one token y.

Discovery

Under which condition is the path $b\,y$ more probable than every path starting with a?



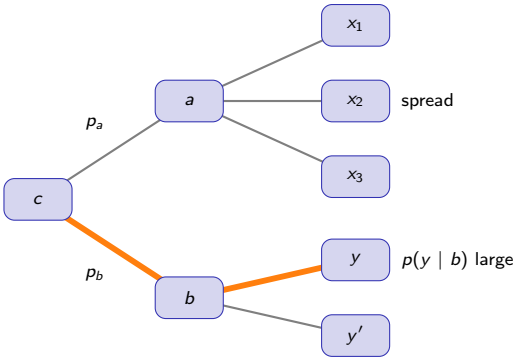
The first token of the most probable continuation can have the smaller probability.

Greedy Path Against the Most Probable Path

Two first tokens a, b with $p_a > p_b$. After a, the mass is spread over many tokens; after b, most of it is on one token y.

Discovery

Under which condition is the path $b\,y$ more probable than every path starting with a?
Answer: when $p_b\, p(y \mid b) > p_a \max_x p(x \mid a)$. Under greedy decoding, b is dropped at the first step. Under beam search with $B \geq 2$, b is kept, and $b\,y$ is the result.



The first token of the most probable continuation can have the smaller probability.

Checkpoint: Beam Width, Search Result

Checkpoint problem

Beam search of width B over V, n steps after a context c.

Question 1: For $n = 2$, which smallest B gives the most probable two-token continuation after every context?

Checkpoint: Beam Width, Search Result

Checkpoint problem

Beam search of width B over V, n steps after a context c.

Question 1: For $n = 2$, which smallest B gives the most probable two-token continuation after every context?

Answer: $\mathbf{B} = |\mathbf{V}|$: every first token is kept, all $|V|^2$ paths are compared. With a smaller B, the best path can start with a dropped token.

Question 2: At the end, which is larger: $\log P$ at width B, or $\log P$ at width $B + 1$?

Checkpoint: Beam Width, Search Result

Checkpoint problem

Beam search of width B over V, n steps after a context c.

Question 1: For $n = 2$, which smallest B gives the most probable two-token continuation after every context?

Answer: $\mathbf{B = |V|}$: every first token is kept, all $|V|^2$ paths are compared. With a smaller B, the best path can start with a dropped token.

Question 2: At the end, which is larger: $\log P$ at width B, or $\log P$ at width $B + 1$?

Answer: either one: at width $B + 1$, another set can be kept at an intermediate step, without the result of width B.

With a wider beam, more paths are compared, and the result can have a smaller $\log P$.

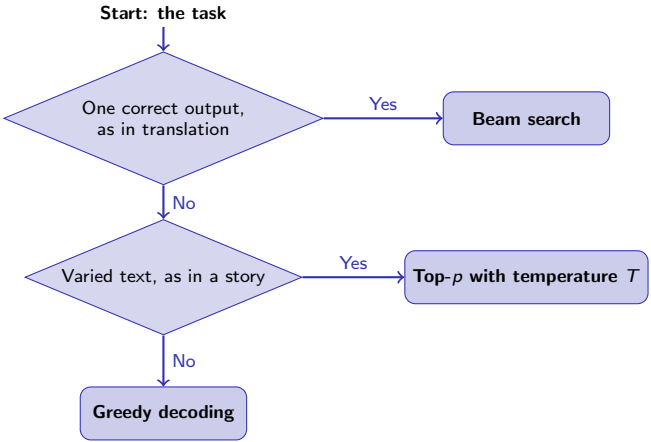
Five Decoding Rules in Six Aspects

Six aspects of the five decoding rules, at one step:

Aspect	Greedy	Temperature	Top-k	Top-p	Beam search
Kind	search	sampling	sampling	sampling	search
Parameter	none	T	k	π	B
One text per prompt	yes	no	no	no	yes
Kept at a step	1 token	all of V	k tokens	$k^*(\pi)$ tokens	B paths
Number kept depends on p	no	no	no	yes	no
Equal to greedy decoding at	every setting	$T \to 0$	$k = 1$	$\pi \to 0$	$B = 1$

Greedy decoding is a limit of each of the four other rules.

From the Task to a Decoding Rule



The decoding rule is chosen from the task, before any text is generated.

Worked Example: GPT-2 Small

- ✓ 1 Next-Token Distribution
- ✓ 2 Greedy Decoding
- ✓ 3 Temperature
- ✓ 4 Top-k, Top-p
- ✓ 5 Beam Search
- ▶ **6 Worked Example: GPT-2 Small**
  Compute each rule on the numbers of GPT-2 small after one prompt.

✓ finished ▶ next under each part ahead: its learning goal

Ten Likeliest Tokens After the Austen Prompt

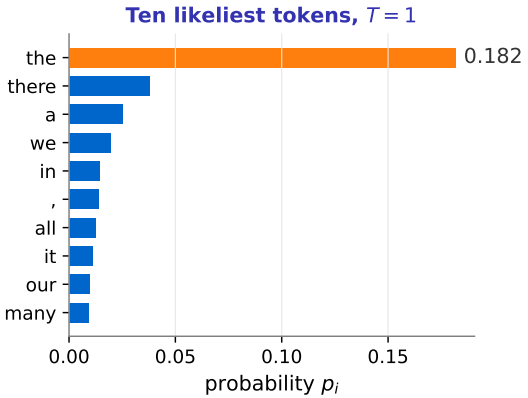
Definition

The **Austen prompt** is the context *It is a truth universally acknowledged that*, the first words of *Pride and Prejudice* (1813). GPT-2 small has $|V| = 50{,}257$ tokens.

Worked: the five likeliest next tokens at $T = 1$:

Token	Score z_i	p_i
`the`	−87.547	0.182
`there`	−89.120	0.038
`a`	−89.514	0.025
`we`	−89.788	0.019
`in`	−90.090	0.014
Five tokens		0.278

Worked: greedy decoding gives `the`; under top-p with $\pi = 0.9$, 3,784 tokens are kept.



After the Austen prompt, the likeliest token has probability 0.182, and 3,784 tokens have a mass of 0.9.

Ratio of the Two Likeliest Tokens at T = 0.5, 1, 2

Step 1: the scores of the two likeliest tokens after the Austen prompt

Token	Score z_i
`the`	-87.547
`there`	-89.120
$\Delta = z_{\text{the}} - z_{\text{there}}$	1.573

Under temperature T, the ratio is $p_{\text{the}}^{(T)} / p_{\text{there}}^{(T)} = e^{\Delta/T}$.

Step 2: the ratio at three temperatures

$$e^{1.573/0.5} = 23.2, \quad e^{1.573/1} = 4.82, \quad e^{1.573/2} = 2.20$$

Reading the results:

- $p_{\text{the}} = 0.885$ at $T = 0.5$, 0.182 at $T = 1$, 0.005 at $T = 2$
- $H = 0.69$, 6.38, 9.88 nats, against $\log |V| = 10.82$
- At every T, `the` is the likeliest token

At $T = 0.5$, `the` is 23.2 times as probable as `there`; at $T = 2$, 2.20 times.

Top-k Mass, Top-p Size After Three Prompts

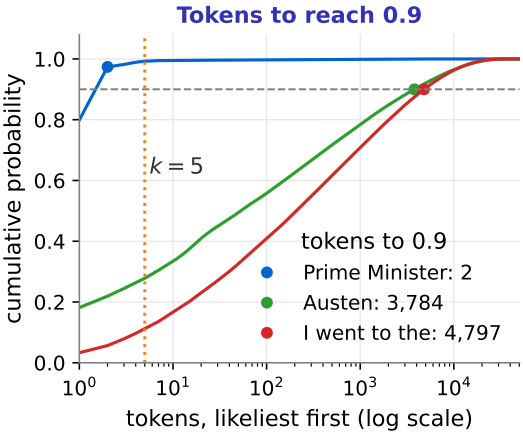
Definition

The **peaked prompt** is *The Prime Minister of the United*; the **flat prompt** is *I went to the*.

Worked: three prompts, top-k mass M_5, top-p size $k^*(0.9)$:

Prompt	$p_{\max}$	Mass M_5	Size $k^*(0.9)$
peaked	0.799	0.992	2
Austen	0.182	0.278	3,784
flat	0.033	0.112	4,797

Worked: under top-k with $k = 5$ after the flat prompt, `hospital` has $q = 0.033/0.112 = 0.295$; under top-p, 4,797 tokens are kept.



At $k = 5$, 0.992 of the mass is kept after the peaked prompt, 0.112 after the flat one.

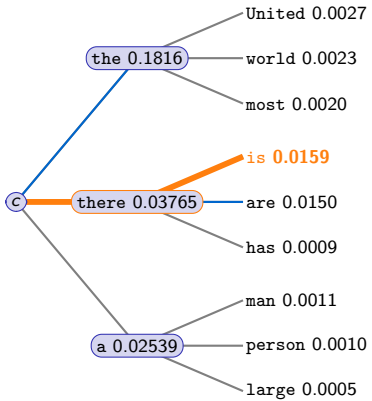
Nine Two-Token Paths After the Austen Prompt

Definition

After the Austen prompt, a **path** is a continuation of two tokens, with the probability $p(w_1 \mid c)\, p(w_2 \mid c, w_1)$. The tree has nine paths: three first tokens, three second tokens after each.

Path	$p(w_1)$	$p(w_2 \mid w_1)$	P
`the United`	0.1816	0.01472	0.0027
`there is`	0.03765	0.4223	0.0159
`there are`	0.03765	0.3976	0.0150

Worked: $B = 1$ gives `the United`, $0.1816 \times 0.01472 = 0.0027$; $B = 2$ gives `there is`, $0.03765 \times 0.4223 = 0.0159$, and $0.0159/0.0027 = 5.9$.



Leaves: P; blue: kept at $B = 2$; orange: the result.

With $B = 2$, the result is `there is`, 5.9 times as probable as the greedy path `the United`.

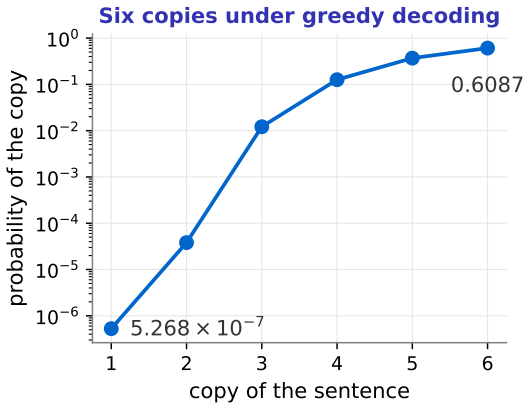
Six Copies of One Sentence Under Greedy Decoding

Definition

A **copy** is one occurrence of the sentence *The United States is a nation of immigrants.* in the greedy continuation of the Austen prompt. Its probability P is the product of its 9 token probabilities.

Copy	Tokens	P
1	1 to 9	5.268×10^{-7}
2	10 to 18	3.812×10^{-5}
3	19 to 27	0.01216
4	28 to 36	0.1263
5	37 to 45	0.3692
6	46 to 54	0.6087

Worked: from copy 1 to copy 6, P grows by $0.6087/(5.268 \times 10^{-7}) = 1.16 \times 10^{6}$.



Under greedy decoding, the probability of the sentence grows with every copy, from 5.268×10^{-7} to 0.6087.

Twenty Tokens After the Austen Prompt Under Five Rules

Five rules, 20 tokens each after the Austen prompt; $\log P$ is the sum of the 20 log-probabilities of GPT-2 small at $T = 1$.

Rule	$\log P$	Continuation
greedy, $B = 1$	-27.77	the United States is a nation of immigrants. The United States is a nation of immigrants. The United
beam, $B = 2$	-27.78	there is no such thing as a ‘good’ or ‘bad’ person. ↩↩ It is
beam, $B = 5$	-25.14	there is no such thing as a “right” or “wrong” way of life. It is
$T = 0.7$, draw 0	-61.53	the best and the brightest minds in the world face incredible pain and surmounting adversity. The best
$T = 1.5$, draw 0	-169.60	WWI and the 1914 Ottoman Imperial Question originally parted rows) Leatherwolves moreapa hamley ancient

Over 20 tokens, $\log P$ **is** -25.14 **under beam search of width 5 and** -169.60 **for a draw at** $T = 1.5$**.**

next_token_dist on the Flat Prompt

```
z = logits("I went to the")                  # 50,257 scores
p = next_token_dist(z, T=1.0)                # T = 1, no draw
q = next_token_dist(z, T=1.0, k=5)           # top-k
r = next_token_dist(z, T=1.0, top_p=0.9)     # top-p
print(word(p.argmax()), round(p.max(), 4))   # hospital 0.0331
print((q > 0).sum(), round(q.max(), 4))      # 5 0.295
print((r > 0).sum())                         # 4797
```

Key outputs:

- `p.max()`: 0.0331, the probability of `hospital`
- `q`: 5 tokens kept, `hospital` at 0.295
- `r`: 4,797 tokens kept at $\pi = 0.9$

Note: $0.0331/0.1122 = 0.295$, the value of top-5 after the flat prompt.

Temperature, top-k, top-p are arguments of one function, `next_token_dist`.

Nucleus Size at Four Temperatures

Discovery Exercise

Open **notebook 01** in Colab, run the setup cell, then the four lines, at four temperatures:

```
z = logits(AUSTEN)
for T in [0.5, 0.7, 1.0, 1.5]:           # four distributions, no draw
    r = next_token_dist(z, T=T, top_p=0.9)
    print(T, (r > 0).sum(), round(r.max(), 3))
```

Questions to answer:

1. What is the size of the nucleus of 0.9 at each T?
2. What is the probability of `the` after the cut at $T = 0.5$, against 0.885 before it?
3. In **notebook 03**, what is the smallest width B with the result `there is`?

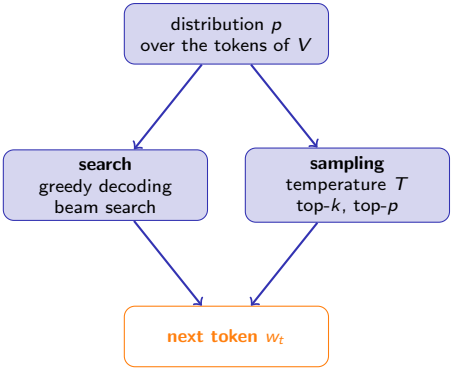
Derivations with checks on GPT-2 small: **notebook 02**.

After the Austen prompt, the nucleus of 0.9 grows with the temperature.

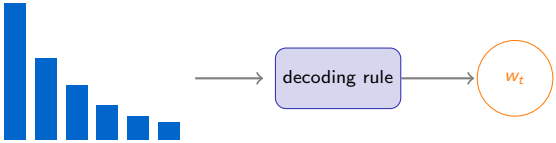
Five Results on Decoding the Next Token

Five results on decoding:

1. Softmax: $p_i = e^{z_i} / \sum_j e^{z_j}$; P of a continuation is a product
2. Greedy decoding: $\arg\max_i p_i$; the probability of a repeated phrase grows with each copy
3. $p^{(T)} \propto e^{z/T}$: greedy decoding at $T \to 0$, uniform at $T \to \infty$
4. Top-k: k tokens fixed; top-p: the mass π fixed
5. Beam search: B paths kept, $B = 1$ greedy; with a wider beam, $\log P$ can be smaller



Each of the five rules is a map from one distribution per step to the next token.



Thank you

Picking the Next Token
Text Analytics, Chapter 7

Appendix

Derivations: entropy, truncation, min-p, repetition penalty

A2: Derivative of the Entropy in T

Given $p_i^{(T)} = e^{\beta z_i}/Z(\beta)$ with $\beta = 1/T$, $Z(\beta) = \sum_j e^{\beta z_j}$; expectations under $p^{(T)}$.

Step 1: $\log p_i = \beta z_i - \log Z$, so the entropy is

$$H = -\sum_i p_i \log p_i = \log Z - \beta\, \mathbb{E}[z].$$

Step 2: $\dfrac{d \log Z}{d\beta} = \mathbb{E}[z]$ and $\dfrac{d\, \mathbb{E}[z]}{d\beta} = \mathrm{Var}(z)$, so

$$\frac{dH}{d\beta} = \mathbb{E}[z] - \mathbb{E}[z] - \beta\, \mathrm{Var}(z) = -\beta\, \mathrm{Var}(z).$$

Step 3: with $d\beta/dT = -1/T^2$, the derivative in T is

$$\frac{dH}{dT} = \frac{\mathrm{Var}_{p^{(T)}}(z)}{T^3} \;\geq\; 0.$$

The entropy of $p^{(T)}$ grows with T at the rate $\mathrm{Var}(z)/T^3$.

A3: Divergence of a Truncation

Given a kept set S with mass $M = \sum_{i \in S} p_i$; $q_i = p_i / M$ on S, $q_i = 0$ elsewhere. Top-k, top-p, min-p are truncations of this form.

Step 1: on S, $q_i / p_i = 1/M$, so

$$\mathrm{KL}(q \,\|\, p) = \sum_{i \in S} q_i \log \frac{q_i}{p_i} = \log \frac{1}{M} \sum_{i \in S} q_i = -\log M.$$

Step 2: under top-p, $M \geq \pi$, so $\mathrm{KL}(q \,\|\, p) \leq -\log \pi$ after every prompt.

Step 3: under top-k, the k largest of $|V|$ probabilities have a mean of at least $1/|V|$, so $M_k \geq k/|V|$ and $\mathrm{KL}(q \,\|\, p) \leq \log(|V|/k)$.

The divergence of a truncation is $-\log M$; it depends on the kept mass alone.

A4: Bound on the Number of Tokens Under Min-p

Definition

Under **min-p** (Nguyen et al., 2024), the kept set is $S = \{\, i : p_i \geq p_{\text{base}}\, p_{\max} \,\}$, with $0 < p_{\text{base}} < 1$.

Step 1: each kept token other than the likeliest has $p_i \geq p_{\text{base}}\, p_{\max}$; together these tokens have a mass of at most $1 - p_{\max}$.

Step 2:

$$(|S| - 1)\, p_{\text{base}}\, p_{\max} \leq 1 - p_{\max} \;\Rightarrow\; |S| \leq 1 + \frac{1 - p_{\max}}{p_{\text{base}}\, p_{\max}}.$$

Interpretation: a peaked p, with $p_{\max}$ near $1 \to$ few tokens; a flat $p \to$ the bound grows as $1/p_{\max}$.

Under min-p, the number of kept tokens is at most $1 + (1 - p_{\max})/(p_{\text{base}}\, p_{\max})$.

A5: Repetition Penalty, Offset of the Scores

Definition

Under the **repetition penalty** $\theta > 1$ (Keskar et al., 2019), the score of each token already in the text is changed to

$$z_i' = \begin{cases} z_i/\theta, & z_i > 0, \\ \theta\, z_i, & z_i \leq 0. \end{cases}$$

Step 1: $\mathrm{softmax}(z + a) = \mathrm{softmax}(z)$ for every constant a: p is the same at every offset of the scores.

Step 2: the penalty depends on the offset: for $z_i < 0$, the score is lower by $(\theta - 1)\,|z_i|$, a difference that grows with $|z_i|$.

Interpretation: two score vectors with one softmax p can give two different distributions after the penalty.

The repetition penalty depends on the offset of the scores; the softmax is the same at every offset.